Unsupervised semantic segmentation of radar sounder data using contrastive learning

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ABSTRACT

Radar Sounders (RSs) are active sensors widely used for planetary exploration and Earth observation that probe the subsurface in a non-intrusive way by acquiring vertical profiles, called radargrams. Radargrams contain information on subsurface geology and are analyzed with neural networks for segmentation and target detection. However, most of these methods rely on supervised training, which requires a large amount of labeled data that is hard to retrieve. Hence, a need emerges for a novel method for unsupervised radargram segmentation. This paper proposes a novel method for unsupervised radargram segmentation by analyzing semantically meaningful features extracted from a deep network trained with a contrastive logic. First, the network (encoder) is trained using a pretext task to extract meaningful features (query). Considering a dictionary of possible features (keys), the encoder training loss can be defined as a dictionary look-up problem. Each query is matched to a key in a large and consistent dictionary. Although such a dictionary is not available for RS data, it is dynamically computed by extracting meaningful features with another deep network called the momentum encoder. Secondly, deep feature vectors are extracted from the encoder for all radargram pixels. After the feature selection, the feature vectors are binarized. Since pixels of the same class are expected to have similar feature vectors, we compute the similarity between the feature vectors to generate a cluster of pixels for each class. We applied the proposed method to segment radargrams acquired in Greenland by the MCoRDS-3 sensor, achieving good overall accuracy.

Keywords: Unsupervised semantic segmentation, Contrastive learning, Image processing, Radar sounder, Subsurface sensing

1. INTRODUCTION

The cryosphere plays a significant role in the Earth climate system, being a sensitive and informative indicator of climate change. However, climate change impact on the dramatic evolution of the cryosphere is still understudied and requires more investigation.[1] Hence, direct measurements of the cryosphere from the surface to the basal interface are required to understand the cryosphere geology and processes better. Direct measurements can be acquired by radar sounders (RSs) that transmit electromagnetic (EM) waves at a low frequency and a wide band. The EM wave is reflected by the dielectric, thermal and mechanical discontinuities in the subsurface. The backscattered echoes are stored in radargrams, profile images of the subsurface.

Given the large number of available radargrams and those planned for acquisition,[2–5] there is a need for automated methods for RS data analysis. In the literature, numerous automatic methods have been proposed to analyze radargrams[6–9] and to extract information about subsurface structures and processes. These methods mostly segment radargrams to identify the subsurface geological targets, i.e., predict a label for each radargram pixel. These machine learning methods were initially based on handcrafted features manually designed for specific targets and cannot generalize to targets not considered in the design.[7,8] Recently, deep learning has raised great interest as it can automatically extract meaningful features from the data and perform well in several tasks

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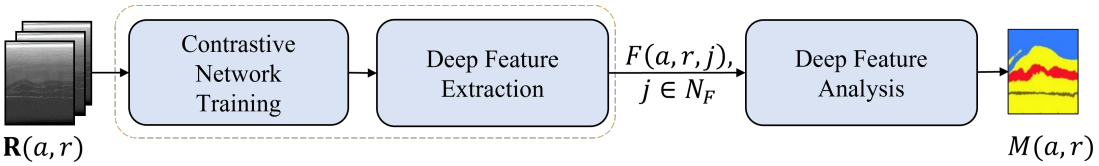


Figure 1: Block diagram of the proposed method.

at the cost of a large training dataset. Deep learning has been applied to radargrams for i) detecting the ice stratigraphy,[10,11] ii) simulating RS images with generative models,[12] iii) target detection,[13] and iv) supervised semantic segmentation.[14–16] Most proposed methods are supervised and require a large training set of data and ground truth couples. In the case of semantic segmentation, a label is required for each pixel considered in the training phase. However, a large labeled dataset, which is required to avoid overfitting, is hard to retrieve in the RS domain. Although several methods use mitigation techniques to reduce the number of labeled data for effective training,[14–16] there is a need for a method to segment radargrams in an unsupervised way.

This work proposes a novel unsupervised method based on deep learning for the semantic segmentation of radar sounder data. The method is based on two steps: first, train a neural network with contrastive logic and a proxy task. Then, we automatically analyze the deep features to segment radargrams. In the first step, the network (encoder) is trained using a pretext task to extract meaningful features (query). Considering a dictionary of possible features (keys), the encoder training loss can be defined as a dictionary look-up problem. Each query is matched to a key in a large and consistent dictionary, which is dynamically computed by extracting meaningful features with another deep network called a momentum encoder. Deep feature vectors are extracted from the encoder for all radargram pixels in the second step. The feature vectors are binarized after a feature selection step to increase sparsity and reduce complexity. Since pixels of the same class are expected to have similar feature vectors, we compute the similarity between the feature vectors to generate a cluster of pixels for each class. We applied the proposed method to segment radargrams acquired in Greenland by the MCoRDS-3 sensor, achieving good overall accuracy.

The structure of the paper is the following: Section 2 proposes the unsupervised method for the semantic segmentation, and Section 3 describes the dataset and the experimental setup and results. Finally, Section 4 concludes the paper and presents the future research directions.

2. PROPOSED UNSUPERVISED SEGMENTATION METHOD

Let us consider radargrams acquired over the central part of ice sheets and caps. We aim to segment the radargram, i.e., assigning a label to each pixel (a, r), to identify geologic subsurface targets. Let the radargram $\mathbf{R}$ be a 2D matrix of N_T traces and N_S samples:

$$\mathbf{R} = \{P_{dB}(a,r) | a \in [1, ..., N_T], r \in [1, ..., N_S]\} \tag{1}$$

where P_{dB} is the power of the backscattered echoes stored in the radargrams in log scale, and (a, r) the azimuth and the range coordinates, respectively. We aim at segmenting radargrams into N_C classes representing meaningful geologic targets and generating a segmentation map $M(a, r)$ that associate a label ω_c to each pixel (a, r). The proposed method consists of three main steps, as shown by Fig. 1. We first train a neural network with a pretext task, i.e., different from the semantic segmentation, with a contrastive loss from Ref. 17. The contrastive loss models the semantic relationship between adjacent pixels and learns an embedding space by pushing close representations for pixels of the same subsurface target and far representations for those of different targets. The trained network is then modified to an autoencoder shape to extract deep semantically meaningful features $\mathcal{F} = F(a, r, j), j \in [0, N_F]$ from unseen radargrams. The features are analyzed to associate each pixel with a label and create the segmentation map $M(a, r)$. As the network extracts more than a hundred features from a radargram, we perform feature selection to discriminate the most informative ones. The features are binarized to increase the sparsity and decrease the complexity of the problem. Finally, assuming that pixels of the same class are expected to have similar feature vectors, we compute the similarity between the feature vectors to generate

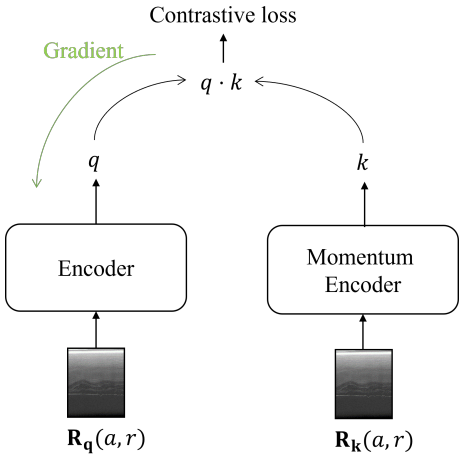


Figure 2: Schematic representation of the Momentum Contrast logic (Fig. adapted from Ref. 17).

a cluster for each class. Considering a set $\mathcal{D}$ of radargrams, we divide it into two independent sets $\mathcal{D}_{tr}$ and $\mathcal{D}_{te}$ that are used for training the network and for extracting and analyzing the features, respectively.

2.1 Contrastive training of the network

The first step aims to extract semantically meaningful features from the radar sounder data in an unsupervised way for the segmentation task. Hence, we propose to train a deep learning model using a pretext task and contrastive learning, similarly to Ref. 17. The pretext task and contrastive logic enable self-supervised training of the network without the need for labeled data.[18] The pretext task enables effective supervised training of the network, thereby defining a closed-form expression for the training loss function to minimize. Finally, contrastive learning aims to train the network to learn general features from the data by teaching it whether two radargrams contain the same target or not.[18]

Here, as a pretext task, we consider the most common self-supervised strategy to input the network with the radargram and its augmented version.[19] Then, using a contrastive loss, the network is trained by knowing that even if the two inputs look different, they contain the same visual information and image the same targets. This forces the latent representations and outputs of similar targets to be similar and those of different targets to be different. As a contrastive logic, we consider the momentum contrast,[17] which transforms the training problem into a dictionary look-up problem. Each input radargram $\mathbf{R}_q$ is associated with a query q, generated by encoding $\mathbf{R}_q$ with a neural network. The query is compared against a large, consistent dictionary of keys $k_i, i \in [0, K]$ to find similar targets. Finally, the query radargram $\mathbf{R}_q$ is matched to the radargram encoded in the dictionary with the lowest distance, and the network learns to differentiate radargrams. To calculate the distance between the query q and a specific key k_i, we consider the dot product and the contrastive function InfoNCE,[20] similarly to Ref. 17, defined as:

$$\mathcal{L} = -\log \frac{\exp(q \cdot k_i/\tau)}{\sum_{i=0}^{K} \exp(q \cdot k_i/\tau)} \tag{2}$$

where τ is a temperature hyper-parameter.[21] The loss aims to estimate the mutual information between q and k_i and forces the representations (features) from a radargram and its augmented version to be close, and those from two unrelated radargrams to be distant. The dictionary should be large, i.e., covering several positive and negative keys, and as consistent as possible. Positive keys encode radargrams with similar visual information and target the query radar image $\mathbf{R}_q$, thus showing high mutual information. Conversely, negative keys encode radargrams with low mutual information with $\mathbf{R}_q$. A consistent dictionary consists of keys computed similarly. Although the dictionary is not available for RS data, it is dynamically computed at each training step by extracting meaningful features with another deep network called the momentum encoder. Such a dictionary does not contain only the mini-batch encoded at that step; it is computed on-the-fly by the updated momentum encoder to increase dictionary consistency. Encoded data are stored and progressively replaced by removing old,

outdated encoded mini-batches that are less consistent with the new keys. Given the use of a large dictionary, it is impossible to update the momentum encoder with back-propagation, as the gradient should propagate through the queue. Authors in Ref. 17 propose to update the momentum encoder using a momentum coefficient $m \in [0, 1]$ defined as:

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q \tag{3}$$

where θ_k represents the parameters of the momentum encoder and θ_q those of the other encoder. θ_q are updated considering back-propagation, while θ_k are updated more smoothly according to coefficient m.

Given the training, the architecture of the two encoders is the same. We propose that each encoder is made of an autoencoder backbone followed by a contracting path. The backbone has an autoencoder architecture to extract features with the same spatial dimension as the input radargrams. The contracting path extracts the query and the keys for momentum-contrast training. As a contracting path, we propose five convolutional blocks to gradually reduce the feature dimension. Each block consists of a 3x3 convolutional layer, a normalization layer, a ReLu activation function, a 1x1 convolutional layer, a normalization layer, and a ReLu activation. A 2x2 max-pooling layer follows each block to reduce the dimension of the extracted features. We apply an adaptive average pooling layer to further reduce the feature dimension by adaptively averaging the input features, followed by a flatten layer that reshapes the input into a 1D tensor. Then, similarly to Ref. 22, we combine the latent 1D features with a multi-layer perceptron (MLP) head, up to dimensionality N_{MoCo}. Conversely, to[17] that concatenates a single fully connected layer head, we propose a 3-layer MLP head (2 hidden layers with ReLU). The output of this contracting path is used for extracting the queries q and the dictionary keys k (defined over $\mathbb{R}^{N_{\text{MoCo}}}$), which are used for training both encoders. Similar to Ref. 15, we propose integrating instance normalization in the backbone and the contracting path instead of batch normalization. Instance normalization performs better when the training set lacks high variability,[23] as is often the case with the radar sounder.

Once both networks are trained on $\mathcal{D}_{tr}$, we process radargrams from $\mathcal{D}_{te}$ during training to extract semantically meaningful features from the backbone (autoencoder) of the encoder.

2.2 Deep feature analysis

For each radargram, we extract a set N_F features $F(a, r, j), j \in [1, N_F]$ to process and generate a segmentation map $M(a, r)$, where each pixel (a, r) is associated with a label $\omega_c, c \in [1, N_C]$. Given the large number of features extracted by the autoencoder (over 100), we propose selecting only the most informative and diverse features. Similar to Ref. 24, we select the most informative features considering the structural similarity index (SSIM) that analyzes the texture information and measures the perceptual difference between features. A high SSIM value indicates high similarity between features, while a low value indicates low resemblance. Hence, we discard features j with high SSIM values and keep those with low values to have $F(a, r, j), j \in [1, N_F^{fs}], N_F^{fs} << N_F$. After min-max normalizing $F(a, r, j)$, we discretize the features to reduce problem complexity and increase feature sparsity while preserving information. The discretization is performed by thresholding $F(a, r, j)$ to obtain $G(a, r, j)$, setting values below 0.5 to 0 and those above 0.5 to 1.

Assuming that each class has a specific binary signature,[13] we check the similarity between the deep feature vector $g_{a,r}$ for each pixel (a, r). We define N_C sample deep vectors $G_r = [g_{a,r}^1, \ldots, g_{a,r}^{N_C}]$ representing the subsurface targets of interest. We compare all the deep feature vectors $g_{a,r}$ to those in G_r, considering the cosine similarity. The cosine similarity measures the similarity between two vectors of an inner product space and can be in the $[0, 1]$. A high cosine similarity value indicates a good similarity between the two vectors, while a low value indicates different vectors. Hence, for each pixel (a, r), we compute the cosine similarity between the deep feature vector $g_{a,r}$ and the sample deep vectors $G_r = [g_{a,r}^1, \ldots, g_{a,r}^{N_C}]$ (see (4)) to have $S_{a,r}(c) = [s_1, \ldots, s_{N_C}]$.

$$s_c = \frac{g_{a,r}^c \cdot g_{a,r}}{\|g_{a,r}^c\| \|g_{a,r}\|} \tag{4}$$

s_c indicates how much the deep vector $g_{a,r}$ is similar to that representing the class c. Thus, using the argmax operator, we can assign to each pixel (a, r) the label ω_c corresponding to the class with the highest similarity:

$$M(a, r) = \underset{c}{\text{argmax}}\{S_{a,r}(c)\} \tag{5}$$

Table 1: Parameters of the radar sounder instrument MCoRDS-3[5] and the RS data geometrical resolutions.

Parameters	Value
Central frequency	195 MHz
Wavelength	1.54 m
Bandwidth	30 MHz
Sampling Frequency	30 MHz
Transmitted Power	1050 W
Aircraft Altitude AGL	$\sim$ 500 m
Range Resolution in Ice	4.3 m
Azimuth Resolution	27.5 m

3. DATASET DESCRIPTION AND EXPERIMENTAL RESULTS

To assess the method's effectiveness, we validate it for the unsupervised segmentation of radargrams acquired in inland Greenland in 2017 that image the ice sheet up to the bedrock. The radargrams are acquired by MCoRDS-3[5] mounted on an aircraft, and the radar details are in Table 1. Here, we consider segmenting radargrams in N_C classes already in the literature,[8] i.e., the ice, the bedrock, and the noise-limited regions (thermal noise, echo-free zone, and free space above the surface). The classes are defined as $\{\omega_c, c \in [1, \ldots, N_C = 3]\}$, where ω_1 is englacial layers, ω_2 is the noise-limited regions, and ω_3 is the bedrock.

3.1 Preprocessing of the radargrams

MCoRDS-3 data are distributed by CReSIS and are already range-compressed, azimuth-focused via synthetic aperture radar (SAR) techniques, and clutter is mitigated.[25] Adjacent radargrams are concatenated by removing overlapping regions. The concatenated data are corrected for fluctuations caused by aircraft platform movements. The power information in the data is log-transformed to approximate additive noise and enhance spatial information.

Dataset generation. Each preprocessed radargram is split into overlapping patches to form the dataset $\mathcal{D}$. The dataset $\mathcal{D}$ is standardized, i.e., normalized by removing the mean and dividing by the standard deviation. Then, $\mathcal{D}$ is divided into two datasets $\mathcal{D}_{tr}$ and $\mathcal{D}_{te}$. $\mathcal{D}_{tr}$ is used to train the encoders. $\mathcal{D}_{te}$ is used for the testing phase, and a reference map is manually generated for each patch so that $\mathcal{D}_{te}$ consists of several couples of patches and the related reference maps. $\mathcal{D}_{tr}$ consists of 346 radargrams that are preprocessed to generate concatenated radargrams with $N_T = 1155133$ traces. $\mathcal{D}_{te}$ consists of 9 radargrams of campaign 20170413_01 that are preprocessed to generate concatenated radargrams with $N_T = 30009$ traces. All the radargrams are divided into overlapping patches of size w and $h = 1280$ in the range and azimuth directions, respectively. h is chosen so that the patches image all the subsurface targets distributed along the range direction. We set w to different values in our experiments, i.e., $w = 256$ or $w = 512$, to increase or decrease the network's field of view. Depending on the value adopted for w, the resulting dataset $\mathcal{D}_{tr}$ consists of 8967 patches ($w = 256$) or 4462 patches ($w = 512$). Data augmentation is performed on the fly on $\mathcal{D}_{tr}$ during training. $\mathcal{D}_{te}$ consists of 40 patches ($w = 256$) or 24 patches ($w = 512$).

Data augmentation. We apply data augmentation to increase the network invariance and robustness with a limited number of training samples.[26] Here, we use standard data augmentation techniques (i.e., affine transformations and horizontal flips) to retrain the geometric properties of geologic structures and generate realistic augmented patches. We apply random horizontal flips and rotations to the original patches to generate augmented patches. Random horizontal flip transformation is applied with a probability $p_f = 0.5$. Random rotation transformation is applied with a probability $p_r = 0.5$, with the rotation degrees in the range $[-\theta_{max}, +\theta_{max}]$, where θ_{max} is the maximum steepness angle of the surface.

3.2 Experimental setup

Architecture setup. As the neural network backbone, any autoencoder can be used. Here, we use a custom network similar to that in Ref. 27, which lacks size-reduction layers. The network is made of $L = 5$ blocks of

i) a convolutional layer with kernel size 3×3 and stride zero that extracts F^l features, ii) a ReLu activation function that introduces non-linearity, and iii) a batch normalization layer to harmonize the feature F^l. After the last block, we add a linear projection layer (a convolutional layer with kernel size 1×1). Finally, the extracted features are instance-normalized, i.e., feature-wise normalized. Note that no layers (e.g., max pooling) reduce the patch width and height; thus, there is no information loss. The contracting path encodes the features to a latent 1D vector of $N_{\text{MoCo}} = 128$ values. We used the SGD optimizer with the weight decay set to $\lambda = 5 \times 10^{-4}$ for optimization. Since we set a small batch size $BS = 16$, we use a relatively high SGD momentum $\mu = 0.9225$ to ensure that a large number of previously seen training samples influence the update in the current optimization step. The initial learning rate is set experimentally to $l_r = 0.0125$ and then updated using a cosine (half-period) learning rate schedule.[28]

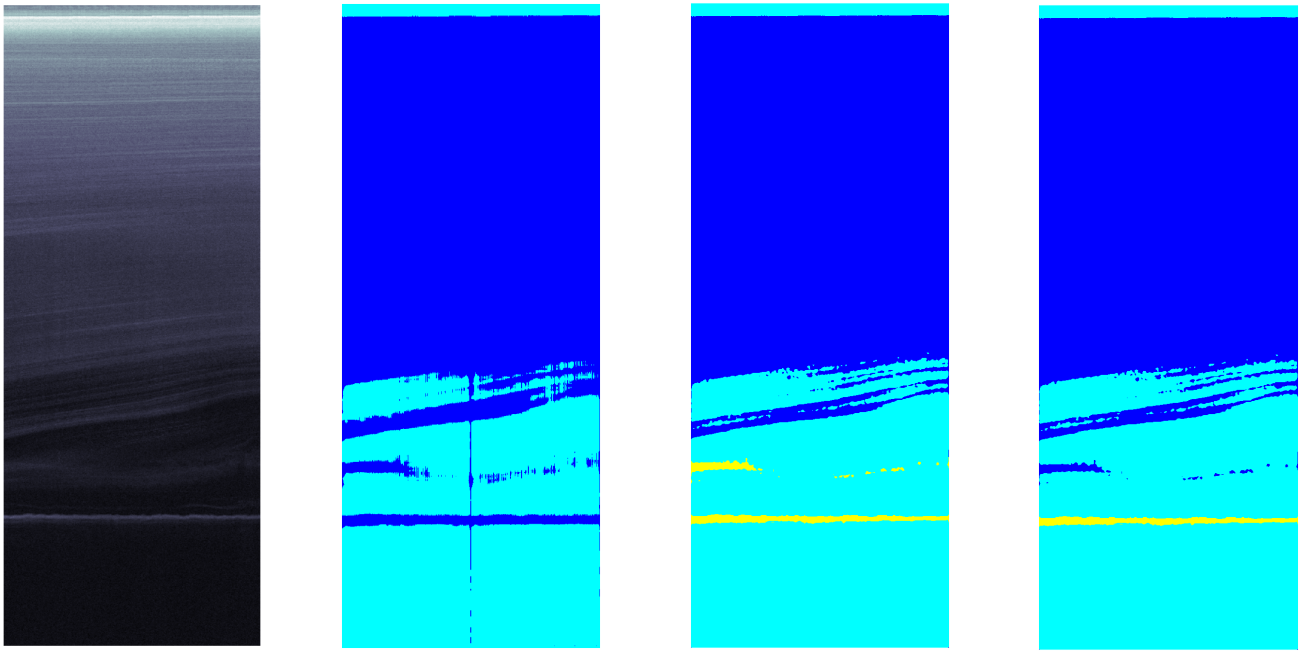
Evaluation metrics. To evaluate the performance of the proposed method, we considered standard segmentation metrics by comparing for each pixel the predicted label with the reference data.[16] Here, we consider the sensitivity (probability that a pixel is classified as the c-th class, given that it belongs to that class) and the specificity (probability that a pixel is not labeled as ω_c, given that it does not belong to that class). Finally, the Overall Accuracy (OA) is defined as the percentage of correctly classified pixels over all the processed pixels.

Comparison baseline. To evaluate the proposed method, we compare it with two unsupervised methods and one supervised method. Similar to the literature,[16] the first method (Case I) extracts N_F features that are then processed by applying softmax so that the features are in the range of $[0, 1]$. N_F is set equal to the number of classes ($N_F = N_C$). Therefore, for each pixel (a, r), considering the deep vector as a posterior probability, we perform the argmax operator to identify the most likely class. The label $\omega_c(a, r)$ is given by the index of the feature in $F(a, r, j)$ that has the maximum value. The second method (Case II) uses the features' variance as the feature selection index, similarly to Ref. 13, rather than the SSIM index. Assuming that more informative features have higher variance, we discard features with low variance. The selected features are then analyzed using the method proposed in Section 2.2. The third method (Case III) performs segmentation in a supervised manner, following the approach in Ref.[16]

3.3 Experimental Results

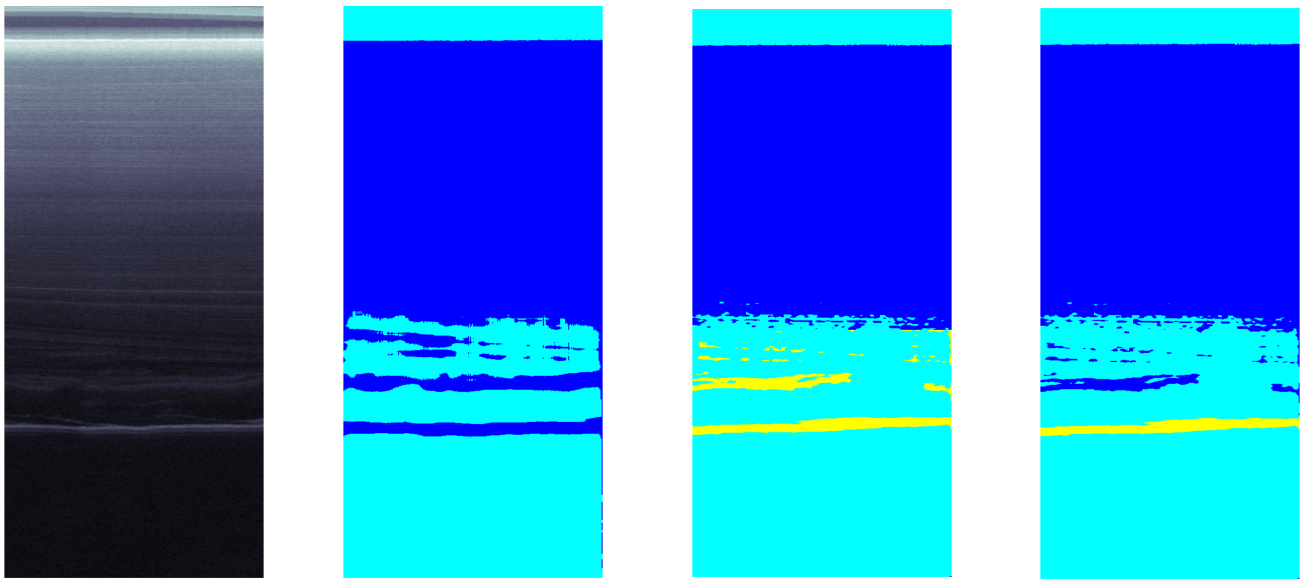
Fig. 3 and 4 show two examples of patches segmented using the proposed method and the baseline methods described in Sec. 3.2 for comparison. On the left is the radargram patch, and on the right are the segmented maps for Case I, Case II, and the proposed method. Table 2 presents the quantitative results of the proposed method and the baseline methods for sensitivity (true positive rate) and specificity (true negative rate). The results show that the proposed method can distinguish the three classes and map the englacial layers, the bedrock, and the noise-limited regions with good accuracy. The sensitivity of the proposed method is always more than 0.86, and the specificity is, on average, 0.83. The proposed method shows slightly better performance in recognizing noise-limited areas than in recognizing englacial layers and bedrock. However, both the proposed and baseline methods have difficulty classifying bedrock pixels.

In general, the proposed method outperforms the baseline ones. Case I shows slightly lower performance across all englacial layers and in the noise-limited regions. However, the bedrock is hardly detected, and the accuracy is typically 0. Note that the only network training with contrastive learning is insufficient to extract meaningful features and perform a segmentation considering the features as posterior probabilities (Case I). Results show that in Case I, the bedrock is missed and those pixels are classified as englacial layers, which is the class with the most similar statistical properties.[8] Since the radiometric and statistical difference between noise-limited regions and signal regions (englacial layers and bedrock) is extremely higher than the difference among the signal classes,[29] the network is learning better to differentiate between the noise-limited and the signal areas than the differences between the subsurface targets (different types of ice and bedrock). Hence, extracting a large number of features increases the network's exposure to features and improves segmentation performance. The performance of the proposed method is slightly better than that of Case II, highlighting the importance of selecting the most informative features and removing the noisy ones. Finally, our fully unsupervised method performs about 10% less than Case III, i.e., the supervised method. This is expected as the method is unsupervised. However, the proposed method achieves almost 90% accuracy without labeled training samples, demonstrating the effectiveness of contrastive training for extracting meaningful features and performing deep feature analysis.



(a) (b) (c) (d)

Figure 3: Examples of the segmented maps of two concatenated patches (a) considering the methods in Case I (b), Case II (c), and the proposed method (d). The englacial layers class is in blue, the bedrock is in yellow, and the noise-limited regions (free space, thermal noise, and echo-free zone) are in light blue.



(a) (b) (c) (d)

Figure 4: Examples of the segmented maps of the patch in (a) considering the methods in Case I (b), Case II (c), and the proposed method (d). The englacial layers class is in blue, the bedrock is in yellow, and the noise-limited regions (free space, thermal noise, and echo-free zone) are in light blue.

Table 2: Segmentation performance of the proposed method and the deep baseline methods (Cases I and II), where ω_1 is the englacial layers, ω_2 is the noise-limited regions, and ω_3 is the bedrock.

Method	Metric	ω_1	ω_2	ω_3	OA
Proposed method	Sensitivity	0.89826	0.90733	0.86090	88.88 %
	Specificity	0.83045	0.87786	0.79742	
Case I	Sensitivity	0.89757	0.76812	0	55.52 %
	Specificity	0.80608	0.87591	1	
Case II	Sensitivity	0.80981	0.88649	0.7561	81.75 %
	Specificity	0.81715	0.71742	0.68622	
Case III[16]	Sensitivity	0.98812	0.97895	0.98483	98.40 %
	Specificity	0.99021	0.98173	0.98318	

4. CONCLUSIONS AND FUTURE WORKS

This paper proposes an unsupervised method for segmenting radargrams into three classes: englacial layers, noise-limited regions, and bedrock. The method accurately segments radargrams in an unsupervised manner without requiring labeled data that are hard to obtain in the RS domain. We train a deep neural network with a pretext task and a contrastive learning logic to extract deep, meaningful features. The deep features are then used to label radargram pixels and generate a segmentation map. The proposed method is tested on radargrams acquired by MCoRDS-3 in northern Greenland, achieving good results. The average accuracy across the three classes is above 88% and outperforms the unsupervised baseline. The comparison with the unsupervised Case I and II highlights the importance of extracting many features and performing effective feature selection to achieve better segmentation maps. The comparison to the supervised Case III shows that the proposed method performs well despite not using labeled training samples.

As future work, we plan to validate the proposed method using additional datasets collected in other areas, e.g., Antarctica and Greenland. We also plan to test the proposed method on a dataset of images of other cryosphere targets, e.g., ice shelves, which are critical for monitoring climate change.

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